Operating Systems – Midterm
April 15, 2014

Total: 100 points

1. Hardware devices signal interrupts when important events occur (e.g., hard-disk data are available):
   (a) (5 points) Show the advantage of interrupt-based operations over busy-waiting.
   (b) (10 points) Describe the process of hardware interrupt handling. You **need to** mention ***device controller, interrupt-request line, state saving, interrupt vector, interrupt-specific handler***.
   (c) (5 points) Explain the benefit of using interrupt vector.

2. (5 points) What is the difference between asymmetric multiprocessing and symmetric multiprocessing?

3. (5 points) How do operating systems support new devices?

4. System calls:
   (a) (10 points) Describe how a system call is invoked using an API function? You **need to** mention ***software interrupt, dual-mode, system call number (index)***, and ***table of code pointer***.
   (b) (5 points) Explain why programmers prefer programming according to API functions rather than invoking system calls directly.
   (c) (5 points) Describe a method to pass system call parameters to the operating system.

5. (5 points) A process in memory generally consists of four major components. What are the components and what is the use of each component?

6. Multithread Models:
   (a) (5 points) Explain the problem of many-to-many thread model without using scheduler activations.
   (b) (10 points) Show how scheduler activations work when a user thread is making a blocking system call. You **HAVE TO** mention ***upcall handler, scheduler***, and ***thread library***.

(b) (10 points) Show your solution satisfies the three requirements of the critical section solution.

(c) (5 points) What is the problem if we simply release the lock in the exist section?

3. (10 points) Deadlocks – consider a system consisting of m resources of the same type, being shared by n processes. Resources can be requested and released by processes only one at a time. Show that the system is deadlock free if the following two conditions hold:

- The maximum need of each process is between 1 and m resources.
- The sum of all maximum needs is less than $m + n$.

4. Deadlocks:

(a) (10 points) **Explain** the four necessary conditions of a deadlock.

(b) (5 points) Can an **UNSAFE** system back to a **SAFE** state? Explain you answer.

(c) (10 points) Consider the following snapshot of a SAFE system; can a request (0,4,2,0) from p_1 be granted? Show your answer using the banker's algorithm.

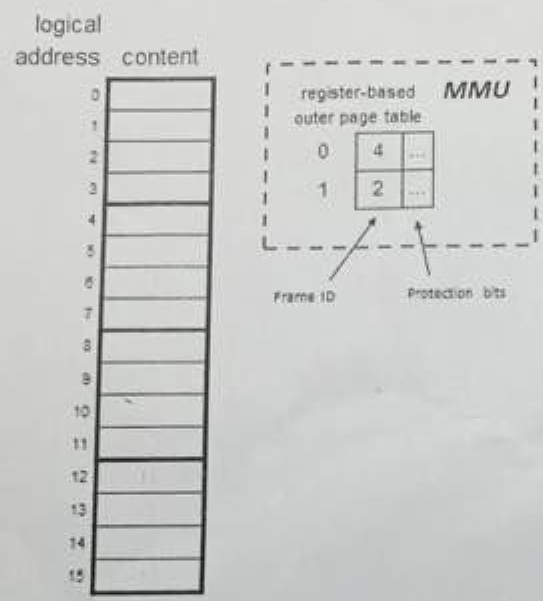
	Allocation				Max			
	A	B	C	D	A	B	C	D
P_0	0	0	1	2	0	0	1	2
P_1	1	~~0~~ 4	~~0~~ 2	0	1	7	5	0
P_2	1	3	5	4	2	3	5	6
P_3	0	6	3	2	0	6	5	2
P_4	0	0	1	4	0	6	5	6

Available			
A	B	C	D
1	5	2	0

5. Paging – Consider a paging system that supports 4-bit logical address and has 32 bytes physical memory. The size of a page is 4 bytes.

(a) (5 points) What is the size of the page table if each table entry costs 2 bytes (the first byte for frame number & the second byte for protection bits)?

(b) (10 points) The following figure shows the physical memory content and the outer page table (two-level paging) of a process. Fill the logical content of the process.



physical address / content

physical address	content
0	0
1	3
2	2
3	5
4	7
5	0
6	2
7	6
8	0
9	3
10	7
11	5
12	4
13	1
14	7
15	3
16	3
17	4
18	5
19	6
20	8
21	10
22	16
23	15
24	2
25	4
26	6
27	8
28	11
29	7
30	9
31	13

6. (10 points) Virtual Memory – Consider the following page reference string:

   1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6.

   How many page faults would occur for the following replacement algorithms, assuming **four** frames are available?

   - LRU replacement
   - Optimal replacement